

DAMA-NET: DISCREPANCY-AWARE FEATURE FUSION AND MULTI-SCALE AGGREGATION NETWORK FOR POLYP SEGMENTATION

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ABSTRACT

Colorectal polyp segmentation is essential for computer-assisted colonoscopy and early colorectal cancer screening, yet remains challenging due to irregular shapes, weak boundaries, low contrast, and complex mucosal backgrounds. Existing methods struggle to coordinate local structural details with long-range contextual information, while fixed-scale modeling limits their adaptability to diverse polyp appearances. To address these challenges, we propose DAMA-Net, a discrepancy-aware feature fusion and multi-scale aggregation network for robust polyp segmentation. A discrepancy-aware feature fusion encoder adaptively combines convolutional local representations with bidirectional RWKV contextual features to enhance local-global feature fusion. Meanwhile, a multi-scale aggregation block dynamically integrates features from different receptive fields to strengthen multi-scale contextual representation. Extensive experiments on four public datasets demonstrate the effectiveness of DAMA-Net. It achieves average Dice and IoU scores of 92.7% and 86.1%, surpassing the strongest competitor by 3.4% and 5.3%, respectively, while requiring only 7.29M parameters and 10.81G FLOPs.

Index Terms— Polyp segmentation, RWKV, discrepancy-aware feature fusion, multi-scale aggregation.

1. INTRODUCTION

Colorectal cancer is one of the most common malignant tumors worldwide, and most colorectal cancers originate from precursor polyps. Accurate polyp segmentation in colonoscopy images is therefore important for early screening and clinical assessment. However, manual annotation is time-consuming and subjective, while segmentation performance is often affected by irregular shapes, weak boundaries, low contrast, illumination variations, and complex mucosal backgrounds.

Deep learning has significantly advanced polyp segmentation through CNN-based, Transformer-based, and state-space-based architectures. CNN-based methods, such as PraNet [1], CAFE [2], and UACANet [3], achieve strong local feature extraction but are limited in long-range dependency modeling. Transformer-based approaches, including TransUNet [4], Polyp-PVT [5], and Swin-UNet [6], enhance global interaction but often introduce high computational costs and may weaken fine-grained structural details. Recently, state-space models such as VM-UNet [7] and CMUNeXt [8] have provided efficient solutions for contextual modeling. Nevertheless, effectively coordinating local structures and global contexts remains challenging for complex polyp images.

To address these challenges, we propose DAMA-Net, a discrepancy-aware feature fusion and multi-scale aggregation network for polyp segmentation. The DFF encoder integrates convolution-based local features with bidirectional RWKV [9] contextual features for local-global feature fusion, while the MA block aggregates multi-scale features from different receptive fields. Extensive experiments on four public datasets, including CVC-300, CVC-ClinicDB, ETIS, and Kvasir-SEG, demonstrate that DAMA-Net outperforms representative CNN-, Transformer-, and RWKV-based methods. On the ETIS dataset, DAMA-Net achieves 94.7% Dice and 90.0% IoU, surpassing the strongest competitor by 2.7% and 4.9%, respectively. Moreover, DAMA-Net requires only 7.29M parameters and 10.81G FLOPs, achieving a better accuracy-complexity trade-off.

2. METHOD

Overall Architecture. The overall architecture of DAMA-Net is illustrated in Fig. 1. Given an input colonoscopy image, a lightweight stem module first extracts initial features. The features are then progressively encoded through four stages with the proposed DFF encoder. By combining convolution-based local extraction and bidirectional RWKV contextual modeling, the DFF encoder enables adaptive local-global interaction. At the bottleneck stage, the MA block integrates features from different receptive fields to enhance contextual representation. Finally, a U-shaped decoder restores spatial

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This work was supported by the Joint Fund Cultivation Project of the Heilongjiang Provincial Natural Science Foundation under Grant JJ2026PL0438.

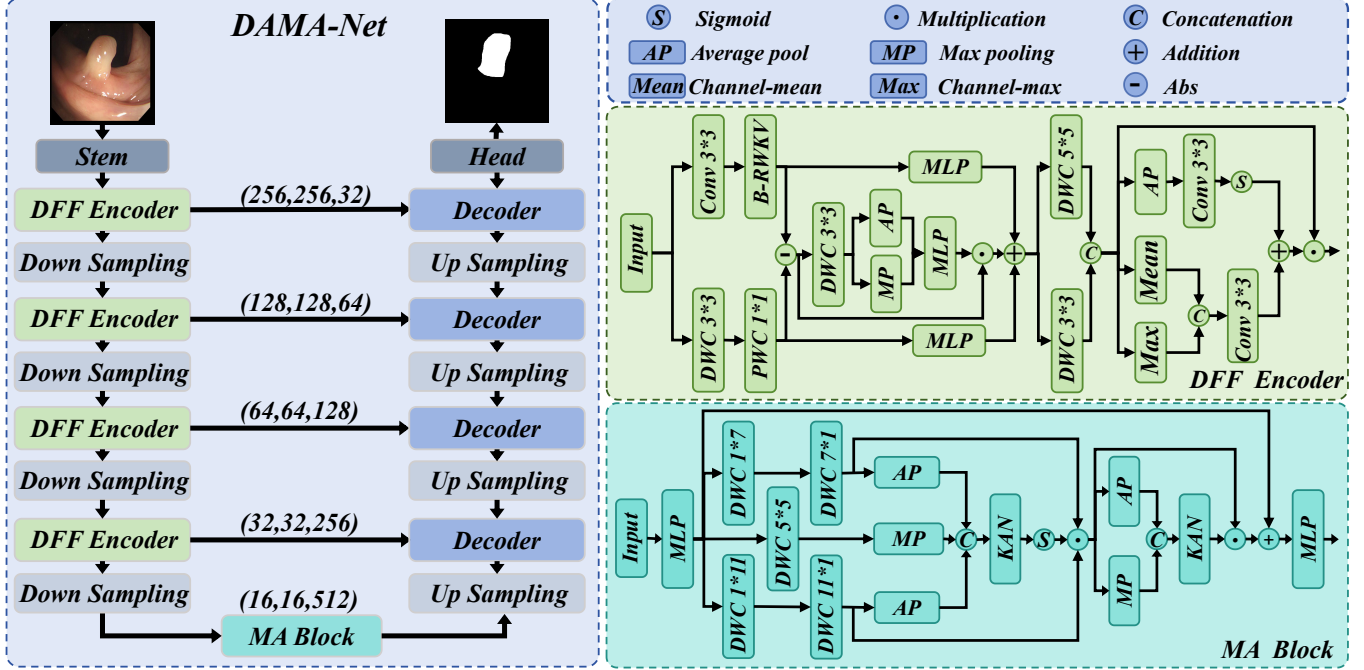


Fig. 1. Overview of DAMA-Net. Input images are processed by a Stem module and sequential DFF encoders for hierarchical features, enhanced by the MA block, then progressively upsampled and fused with encoder features via the decoder.

resolution and fuses multi-level features through skip connections to generate the segmentation mask.

Discrepancy-aware Feature Fusion Encoder. The DFF encoder is designed to adaptively fuse local spatial features and long-range contextual representations for effective polyp feature learning. Given an input feature map $X_{in} \in R^{C \times H \times W}$, where C , H , and W denote the channel number, height, and width, respectively, the input is first processed by two parallel branches.

The contextual branch applies a 3×3 convolution followed by bidirectional RWKV (B-RWKV) to model long-range dependencies, while the local branch employs depthwise convolution and pointwise convolution to capture fine-grained spatial structures. Specifically, B-RWKV processes feature sequences along two complementary spatial directions and aggregates the generated representations. The outputs of the two branches are denoted as local feature X_l and contextual feature X_g , respectively.

To characterize the inconsistency between local structures and global contextual responses, the DFF encoder calculates the difference between the two representations:

$$F_d = |X_l - X_g|, \quad (1)$$

where F_d represents the feature discrepancy between the two branches. The discrepancy feature is further processed by depthwise convolution, followed by average pooling and max pooling. The pooled features are concatenated and fed into

an MLP to generate adaptive weights $\alpha(F_d)$. Meanwhile, the local and contextual features are independently transformed by MLP layers and adaptively fused as:

$$F_{fuse} = \alpha(F_d) \odot \text{MLP}(X_l) + (1 - \alpha(F_d)) \odot \text{MLP}(X_g). \quad (2)$$

The fused feature is further refined through multi-scale convolution operations. Specifically, two depthwise convolution branches with kernel sizes of 3×3 and 5×5 are applied to capture complementary spatial patterns. The outputs of the two branches are concatenated to obtain the multi-scale representation F_m . Subsequently, F_m is processed by two parallel recalibration branches. The first branch extracts spatial responses by computing channel-wise mean and max statistics, followed by concatenation and a 3×3 convolution. The second branch generates channel responses through average pooling, convolution, and sigmoid activation. The outputs of the two branches are summed to generate the recalibration weight, which is then applied to F_m through element-wise multiplication:

$$F_{out}^D = F_m \odot (\mathcal{A}_s(F_m) + \mathcal{A}_c(F_m)), \quad (3)$$

where $\mathcal{A}_s(\cdot)$ and $\mathcal{A}_c(\cdot)$ denote the spatial and channel recalibration branches, respectively.

Multi-scale Aggregation Block.

The MA block is designed to aggregate contextual information from different receptive fields and enhance the representation of polyps with diverse scales and appearances.

Given the deepest encoder feature $X_{\text{in}}^A \in R^{C \times H \times W}$, where C , H , and W denote the channel number, height, and width, respectively, the input feature is first projected through an MLP layer and then processed by three parallel depthwise convolution branches with different receptive fields.

Specifically, the first branch employs asymmetric depthwise convolutions with 1×7 and 7×1 kernels, the second branch adopts a 5×5 depthwise convolution, and the third branch utilizes 1×11 and 11×1 depthwise convolutions. These branches capture contextual information from different spatial ranges. The outputs of the three branches are denoted as:

$$F_i = \Phi_i(X_{\text{in}}^A), \quad i = 1, 2, 3, \quad (4)$$

where $\Phi_i(\cdot)$ represents the corresponding multi-scale convolution branch.

After multi-scale feature extraction, the first and third branch features are processed by average pooling, while the second branch feature is processed by max pooling. The resulting statistics are concatenated through a concatenation operation $\text{Cat}(\cdot)$ and transformed by a KAN layer $\text{KAN}(\cdot)$, followed by a sigmoid activation $S(\cdot)$ to generate adaptive scale weights. These weights are then applied to recalibrate the first and third branch features through element-wise multiplication. The enhanced multi-scale representation is formulated as:

$$F_{ms} = S(\text{KAN}(\text{Cat}(F_1, F_2, F_3))) \odot (F_1 + F_3), \quad (5)$$

The recalibrated feature is further processed through another aggregation stage. Specifically, average pooling and max pooling are applied to extract complementary contextual statistics, and the pooled features are concatenated and transformed by KAN to generate adaptive weights. The resulting weights are used to refine the multi-scale representation through element-wise multiplication. Finally, the refined feature is fused with the residual feature and passed through an MLP layer to obtain the output representation:

$$F_{\text{out}}^A = \text{MLP}(F_{ms} \odot \mathcal{W}(F_{ms}) + X_{\text{in}}^A), \quad (6)$$

where $\mathcal{W}(\cdot)$ denotes the adaptive weighting operation generated by KAN.

Through hierarchical multi-scale aggregation and adaptive feature recalibration, the MA block enables the network to capture contextual information from different receptive fields and improve feature representation for polyps with varying scales and shapes.

3. EXPERIMENTS

Datasets. Experiments are conducted on four public polyp segmentation datasets, including CVC-300 [14], ETIS [15],

Table 1. Quantitative comparison on four polyp segmentation datasets. Dice and IoU are reported in percentage (%). The p-values are obtained by Wilcoxon signed-rank test between each method and DAMA-Net. The best and second-best results are highlighted in bold and underline, respectively.

Model	CVC-300		ClinicDB		ETIS		Kvasir-SEG		p-value
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	
PraNet	88.1	78.7	92.7	86.4	87.6	78.0	84.2	72.8	4e-6
nnUNet	84.8	73.6	<u>93.0</u>	87.0	88.0	81.6	81.5	68.8	1e-6
TransUNet	91.5	83.9	92.7	86.4	89.7	81.3	70.1	54.0	2e-4
PVT-EMCAD [10]	89.8	81.4	<u>93.0</u>	86.9	90.2	82.2	84.1	72.5	7e-5
H2Former [11]	69.7	53.5	89.9	81.7	79.1	65.4	83.6	71.8	2e-15
U-KAN [12]	<u>91.6</u>	<u>84.7</u>	90.5	82.6	<u>90.3</u>	<u>82.3</u>	<u>84.8</u>	<u>73.6</u>	7e-8
U-RWKV [13]	87.7	78.1	91.0	83.4	90.1	82.0	83.9	72.3	4e-6
DAMA-Net	93.4	87.6	96.3	91.0	94.7	90.0	86.2	75.7	–

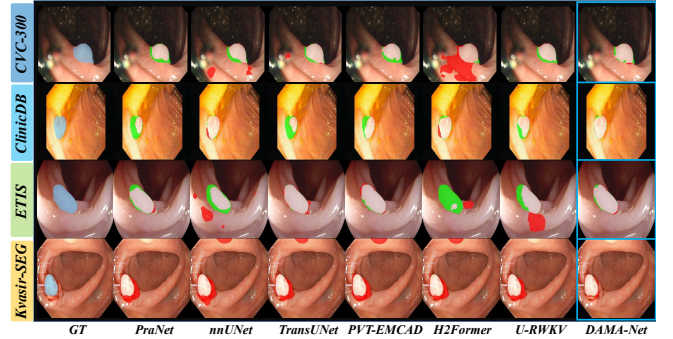


Fig. 2. Qualitative comparison on CVC-300, ClinicDB, ETIS, and Kvasir-SEG. White, red, and green denote true-positive, false-positive, and false-negative regions, respectively.

CVC-ClinicDB [16], and Kvasir-SEG [17]. CVC-300 contains 300 images with clear polyp structures, CVC-ClinicDB includes 612 images with diverse polyp shapes, ETIS provides 196 challenging cases with low contrast, and Kvasir-SEG contains 1000 clinical images with large appearance variations. These datasets provide comprehensive benchmarks under diverse imaging conditions, covering various lesion scales, visual patterns, and segmentation difficulties.

Implementation Details. All experiments are implemented using PyTorch. The input images are uniformly resized to 256×256 pixels. For fair comparison, all datasets are randomly split into training, validation, and test sets with a ratio of 8:1:1 using the same random seed (3407). All compared methods are re-implemented and trained under identical experimental settings. DAMA-Net is optimized using the AdamW optimizer with an initial learning rate of 3×10^{-4} , a batch size of 8, and a weight decay of 1×10^{-4} . The training objective combines Dice loss and cross-entropy loss with equal weighting, and all experiments are conducted on

Table 2. Computational complexity comparison of different polyp segmentation methods. Params, FLOPs, Avg Dice, and Avg IoU are reported in M, G, and percentage (%), respectively.

Method	Params	FLOPs	Avg Dice	Avg IoU
PraNet	30.50	13.77	88.2	79.0
nnUNet	<u>7.85</u>	27.90	76.8	65.3
TransUNet	105.32	76.78	86.0	76.4
PVT-EMCAD	26.76	<u>11.59</u>	<u>89.3</u>	<u>80.8</u>
H2Former	33.69	64.81	80.6	68.1
U-KAN	25.35	21.16	89.3	80.8
U-RWKV	14.86	43.54	88.2	79.0
DAMA-Net	7.29	10.81	92.7	86.1

an NVIDIA RTX 3090 GPU.

Quantitative comparison. Table 1 compares DAMA-Net with representative CNN-, Transformer-, and RWKV-based methods on four public polyp segmentation datasets. DAMA-Net consistently achieves the best Dice and IoU scores across all benchmarks, demonstrating its effectiveness over existing approaches. Specifically, DAMA-Net obtains Dice scores of 93.4%, 96.3%, 94.7%, and 86.2% on CVC-300, ClinicDB, ETIS, and Kvasir-SEG, respectively. On the challenging ETIS dataset, DAMA-Net achieves 94.7% Dice and 90.0% IoU, surpassing the second-best method by 4.4 and 7.7 percentage points, respectively. On ClinicDB, DAMA-Net reaches a Dice score of 96.3%, improving the previous best result by 2.4 percentage points. Furthermore, the Wilcoxon signed-rank tests indicate that the improvements over all competing methods are statistically significant, indicating the reliability of the performance gains. These results show that the proposed DFF encoder and MA block effectively enhance local-global feature fusion and multi-scale contextual representation for accurate polyp segmentation.

Qualitative comparison. As shown in Fig. 2, several competing methods generate incomplete lesion regions or introduce false-positive responses in surrounding mucosal areas. These errors are more noticeable in cases with small lesions, ambiguous boundaries, and irregular shapes. In contrast, DAMA-Net produces predictions closer to the ground truth, with complete lesion coverage and fewer background artifacts, demonstrating effective local-global feature fusion and multi-scale contextual aggregation. Moreover, DAMA-Net maintains consistent segmentation quality across lesion appearances, indicating robustness to challenging segmentation scenarios. These observations further highlight the capability of DAMA-Net to preserve fine-grained structures while suppressing irrelevant background responses.

Complexity analysis. Table 2 compares the computational complexity and segmentation performance of different polyp segmentation methods. DAMA-Net achieves the highest Avg Dice and Avg IoU scores while maintaining the lowest com-

Table 3. Ablation study of different components on CVC-300 and ETIS.

Components			Complexity		CVC-300		ETIS		Average	
Base	DFF	MA	Params	FLOPs	Dice	IoU	Dice	IoU	Dice	IoU
✓			14.86	43.54	87.7	78.1	90.1	82.0	88.9	80.1
✓	✓		9.49	30.50	90.9	83.2	91.6	83.7	91.3	83.5
✓	✓	✓	7.29	10.81	93.4	87.6	94.7	90.0	94.1	88.8

putational complexity among all compared methods. Specifically, DAMA-Net contains only 7.29M parameters and requires 10.81G FLOPs, which are lower than all competing CNN-, Transformer-, and RWKV-based methods. Compared with the Transformer-based TransUNet, DAMA-Net reduces the number of parameters by 93.1% and FLOPs by 85.9%, while improving the Avg Dice from 86.0% to 92.7%. Moreover, compared with nnUNet, which has a comparable parameter scale, DAMA-Net achieves a substantial improvement in Avg Dice from 76.8% to 92.7% with fewer parameters and lower computational cost. These results demonstrate that DAMA-Net achieves an excellent accuracy-complexity trade-off for polyp segmentation.

Ablation study. Table 3 investigates the contributions of DFF and MA based on the U-RWKV baseline. Introducing DFF improves the Dice score from 87.7% to 90.9% on CVC-300 and from 90.1% to 91.6% on ETIS, supporting the effectiveness of local-global feature fusion. With the further addition of MA, DAMA-Net achieves Dice scores of 93.4% and 94.7% on CVC-300 and ETIS, respectively. Compared with the baseline, the complete model improves Dice by 5.7 and 4.6 percentage points, while reducing the complexity to 7.29M parameters and 10.81G FLOPs. These results indicate that DFF and MA provide complementary improvements by enhancing feature representation and contextual aggregation. Moreover, the consistent gains across different datasets demonstrate the effectiveness of DAMA-Net.

4. CONCLUSION

In this paper, we propose DAMA-Net, a discrepancy-aware feature fusion and multi-scale aggregation network for accurate and efficient polyp segmentation. The proposed DFF encoder and MA block effectively enhance local-global feature fusion and multi-scale contextual representation. Extensive experiments on four public datasets demonstrate that DAMA-Net consistently outperforms representative CNN-, Transformer-, and RWKV-based methods, achieving superior segmentation performance with only 7.29M parameters and 10.81G FLOPs. Ablation studies further validate the effectiveness of the proposed components, highlighting the robustness and efficiency of DAMA-Net for challenging polyp segmentation scenarios.

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